Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Schrödinger equation is equivalent to information conservation

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy.lean`
- Library: `UnitaryEntropy`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, the Wigner library files `Density.lean` and `Differentiable.lean` (which imports the rest of `ForPhyslib/QuantumMechanics/Wigner`), and two new `ForMathlib` files:
  `ForMathlib/Analysis/Calculus/Deriv/Star.lean` (in a normed star algebra, the derivative of a path of unitaries through 1 is skew-adjoint, and $U^\dagger U$ is constant when $U' = XU$ with X skew-adjoint),
  `ForMathlib/Analysis/InnerProductSpace/Adjoint.lean` (in finite dimension, $U^\dagger U = 1$ already makes U unitary)
- Wigner library, in `ForPhyslib/QuantumMechanics/Wigner/`: `Density` (density operators, pure states `proj`, equal mixtures `mix`; moved there from this brief), `Basic`, `Normalize`,

`Rigidity`, `Theorem` (Wigner's theorem without bijectivity, after Gehér, now in any dimension, see Section 1.1; uniqueness up to one phase in finite dimension), `Projection` (the same for maps on rank-one projections), `Continuity` (Bargmann's triple product excludes the anti-unitary case near t_0), `Differentiable` (explicit, differentiable choice of the phase). Its `ForMathlib` helpers are `Analysis/Complex/ThreeCircles.lean`, `Analysis/InnerProductSpace/PhaseUnitary.lean` and `Analysis/InnerProductSpace/LinearMap.lean` (conjugate-linear isometries conjugate the inner product; polarization of $|x\rangle\langle y|$). Coordinate conjugation in an orthonormal basis came from TauCeti (`TauCeti.Analysis.InnerProductSpace.Conjugation`) until the library moved to Hilbert bases.

Setting. E is a finite-dimensional complex Hilbert space, and operators are continuous linear maps $E \to E$. A density operator (`IsDensity`) is a positive operator of trace 1. The pure state of a unit vector ψ is `proj` $\psi = |\psi\rangle\langle\psi|$ (Mathlib's `rankOne`), and `mix` $\rho\,\sigma = \frac{1}{2}\rho + \frac{1}{2}\sigma$. The von Neumann entropy `vonNeumann` is $\sum -\mu \log \mu$ over the roots μ of the characteristic polynomial. For self-adjoint ρ these are its eigenvalues (`vonNeumann_eq_sum`), and the definition makes invariance under conjugation immediate. `born` $\psi\,\phi = |\langle\psi|\phi\rangle|^2/(\|\psi\|^2\|\phi\|^2)$, and `mixEntropy` $p = H_2((1+\sqrt{p})/2)$ with H_2 Mathlib's `binEntropy`. The Schrödinger equation for a propagator is $U'(t) = (i\hbar)^{-1}H(t)\,U(t)$ with $H(t)$ self-adjoint.

Results.

Brief	Lean
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumann_mix_proj`, `mixEntropy_eq`
Corollary 1, strictly decreasing	`mixEntropy_strictAntiOn` (on $[0,1]$)
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \dots$	`born_eq_iff` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary propagator	`star_mul_self_of_schrodinger`, `ContinuousLinearMap.mem_unitary_of_star_mul_self`
Theorem 2 ($\Rightarrow$), existence for constant H	`hasDerivAt_exp_schrodinger`: $U(t) = \exp(t(i\hbar)^{-1}H)$
Theorem 2 ($\Rightarrow$), DR-INFO	`vonNeumann_unitary_conj`, `IsDensity.unitary_conj`, `schrodinger_conserves_entropy`
Theorem 2 ($\Leftarrow$), Step 1	`vonNeumann_eq_zero_iff`: $S(\rho) = 0$ iff ρ is pure
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves`
Theorem 2 ($\Leftarrow$), Step 2, Wigner's theorem	`Wigner.wigner_rankOne` (from `Wigner.wigner`)
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`Wigner.eventually_exists_unitary`, `PhaseEqOnSphere.false_of_unitary_of_antiunitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)$	`Wigner.exists_unitary_differentiableAt`
Theorem 2 ($\Leftarrow$), Step 3	`exists_hamiltonian` (from `star_eq_neg_of_hasDerivAt_of_mem_unitary`), for $\hbar \neq 0$
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves`

Differences from the brief.

- This part is finite-dimensional; the brief has a separable Hilbert space. The calculus results (`star_mul_self_of_schrodinger`, `exists_hamiltonian`) hold in any Hilbert space for bounded $H(t)$. Corollary 1 and Theorem 2 through Wigner's theorem are redone in any dimension in Section 1.1.

- Corollary 1 is proved without diagonalizing the mixture. $\rho = \frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$ satisfies $\rho^3 = \rho^2 - \frac{1-\lambda}{4}\rho$ (`mix_proj_cubic`), so every eigenvalue is 0 or $(1 \pm \sqrt{\lambda})/2$. On these three points $-\mu \log \mu = \alpha\mu + \beta\mu^2$ for suitable α, β, and $\operatorname{tr}\rho = 1$, $\operatorname{tr}\rho^2 = (1+\lambda)/2$ fix the sum. This also covers the degenerate cases $\lambda = 0$ (double eigenvalue $\frac{1}{2}$) and $\lambda = 1$ (ρ pure).

B EVMIX is used only for equal mixtures, which is all Step 2 needs. For the ($\Rightarrow$) direction, $\rho \mapsto U\rho U^\dagger$ is linear, so it preserves all mixtures.

- DR-INFO is $S(\rho_I) = S(\rho_F) = S(\rho_P)$. The reconstructed past $U^\dagger \rho_F U$ is ρ_I itself, which is also proved.

- The ($\Rightarrow$) direction shows that every propagator solving the equation is unitary. Existence is shown for constant H only; for time-dependent $H(t)$ it is the linear ODE existence theorem, which the brief also assumes.

- `schrodinger_of_preserves` assumes $\Phi_{t_0} = \mathrm{id}$ and differentiability of $t \mapsto \Phi_t(\rho)$ at t_0 only on density operators, one ρ at a time. It concludes that $\Phi_t(|\psi\rangle\langle\psi|) = |U(t)\psi\rangle\langle U(t)\psi|$ for t near t_0, with $U(t)$ unitary for every t, $U(t_0) = 1$ and $U'(t_0) = (i\hbar)^{-1} H U(t_0)$ for a self-adjoint H. This is the Schrödinger equation at t_0, which is what EVDIFF at t_0 gives; the brief's two-time evolution maps give it at every initial time.

- Wigner's theorem is used in its non-bijective form (Gehér), so the bijectivity of the correspondence of pure states via the reconstructed past (Step 1) is not needed.

- Not formalized: $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for mixed ρ. It would need Φ_t to be affine (Kadison), while the brief assumes only equal mixtures, and the conclusion of the brief concerns U only.

Problems found in the brief. No errors. Implicit points:

- The "iff" in Corollary 1 needs $\lambda \in [0, 1]$, where the entropy is strictly decreasing in λ; for other λ, $\sqrt{\lambda}$ and the logarithms are not meaningful.

- Step 3 assumes a norm-differentiable unitary family, which forces a bounded generator. On an infinite-dimensional Hilbert space it therefore covers bounded Hamiltonians only. The brief "Multiple equivalent characterizations of the Schrödinger equation" treats the general case with Stone's theorem.

- In Step 3, $\hbar \neq 0$ is needed to write $A = H/(i\hbar)$.

- Wigner's theorem gives each $U(t)$ only up to a phase, and Step 3 differentiates $U(t)$. A differentiable choice exists. Extend Φ_t to all $|x\rangle\langle y|$ by polarization, so that it sends $|x\rangle\langle y|$ to $|Ux\rangle\langle Uy|$, and fix an orthonormal basis (b_k) and an index i. Then

$$U(t) = \left(\operatorname{Re}\langle b_i|\Phi_t(|b_i\rangle\langle b_i|)\, b_i\rangle\right)^{-1/2} \sum_k \Phi_t(|b_k\rangle\langle b_i|)\ |b_i\rangle\langle b_k|$$

  is the unitary with $\langle b_i|U(t)b_i\rangle > 0$. It is a finite expression in values of Φ_t on pure states, so it is differentiable at t_0 whenever these are.

- "Joint continuity and $\Phi_{t_0} = I$ exclude the anti-unitary case" needs only continuity at t_0 of $t \mapsto \Phi_t(P)$ for three pure states P: the imaginary part of Bargmann's triple product $\mathrm{tr}(\Phi_t(P_1)\Phi_t(P_2)\Phi_t(P_3))$ is $\frac{1}{4}$ at t_0 and would be $-\frac{1}{4}$ for an anti-unitary. In dimension 1 there is nothing to exclude.

1.1 Hilbert spaces of any dimension

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/InfiniteDimensional.lean`
- Library: `UnitaryEntropy`, which now builds every module `UnitaryEntropy.*`
- Imports:
  `ForPhyslib.QuantumMechanics.VonNeumannEntropy`,
  `ForMathlib.Analysis.InnerProductSpace.StrongDerivative`,
  `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison` (eigenvector action of the functional calculus, for the comparison in finite dimension),
  and the finite-dimensional part `UnitaryEntropy` (for `born`, `mixEntropy`, `mix_proj_cubic`, `mix_proj_sq`, `star_mul_self_of_schrodinger` and `exists_hamiltonian`, none of which assumes finite dimension)
- New file `ForPhyslib/QuantumMechanics/DensityOperator.lean`: density operators on a Hilbert space of any dimension, on top of the trace-class operators of PhyslibAlpha (`PhyslibAlpha/ProbabilisticTheory/HilbertSpace/TraceClass/`, from the physlib dependency; it builds against our Mathlib pin). Pure states, convex combinations, conjugation by an isometry, idempotent density operators are pure, and agreement with `IsDensity` in finite dimension.
- New file `ForPhyslib/QuantumMechanics/VonNeumannEntropy.lean`: the von Neumann entropy in $[0, \infty]$, its invariance under conjugation by an isometry, and $S(\rho) = 0$ iff ρ is pure.
- New `ForMathlib` material:
  `ForMathlib/Analysis/InnerProductSpace/PositiveTrace.lean` (the trace of a positive operator in $[0, \infty]$, independent of the Hilbert basis),
  `ForMathlib/Analysis/InnerProductSpace/HilbertBasisMaps.lean` (for a Hilbert basis: a vector is determined by its coordinates, Parseval for their moduli, the diagonal unitary $b_i \mapsto \delta_i b_i$, coordinatewise conjugation, the isometry $b_i \mapsto v_i$ onto an orthonormal family, and $V(V^\dagger x) = x$ when $\|V^\dagger x\| = \|x\|$),
  `ForMathlib/Analysis/InnerProductSpace/StrongDerivative.lean`
  (by Banach–Steinhaus, the strong derivative of an operator family is a bounded operator; for unitaries with $U(t_0) = 1$ it is skew-adjoint).
- The Wigner library `ForPhyslib/QuantumMechanics/Wigner/` (`Basic`, `Normalize`, `Rigidity`, `Theorem`) now works over a Hilbert basis of any Hilbert space. `Theorem` proves Wigner's theorem in any dimension, with isometries, and derives the finite-dimensional theorem from it. `Continuity` excludes the anti-unitary branch in any dimension, without a trace, and derives the finite-dimensional exclusion from it. `Differentiable` adds the differentiable choice of the phase in any dimension. The other files built on the library are unchanged.

Setting. E is a complex Hilbert space of any dimension; separability is not used. `posTrace` $T = \sum_i \langle e_i|Te_i\rangle \in [0,\infty]$ for a Hilbert basis (e_i). For positive T it does not depend on the basis (`posTrace_eq_tsum`): write $\langle e_i|RR^\dagger e_i\rangle = \|R^\dagger e_i\|^2$, expand by Parseval in a second basis and exchange the two sums of nonnegative terms. A density operator (`IsDensityOperator`) is a positive operator that is trace class in the sense of PhyslibAlpha (`IsTraceClass`) with `trace` $\rho = 1$; equivalently, $0 \le \rho$ and `posTrace` $\rho = 1$ (`isDensityOperator_iff`). The entropy is `vonNeumannEntropy` $\rho =$ `posTrace` $\eta(\rho)$ with $\eta(x) = -x\log x$ and $\eta(\rho)$ given by the continuous functional calculus. The spectrum of a density operator lies in $[0,1]$, where $\eta \ge 0$, so $\eta(\rho)$ is positive. `proj`, `mix`, `born` and `mixEntropy` are as above.

Results.

Brief	Lean
QST:H, density operators	`IsDensityOperator`, `isDensityOperator_proj`, `IsDensityOperator.mix`
QST:H, von Neumann entropy	`vonNeumannEntropy`
Agreement in finite dimension	`isDensityOperator_iff_isDensity`, `vonNeumannEntropy_eq_ofReal_vonNeumann`
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumannEntropy_mix_proj`
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \dots$	`born_eq_iff_vonNeumannEntropy` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary conjugation preserves the entropy	`IsDensityOperator.vonNeumannEntropy_conj` (for isometries)
Theorem 2 ($\Rightarrow$), DR-INFO	`IsDensityOperator.conj`, `schrodinger_conserves_vonNeumannEntropy`
Theorem 2 ($\Leftarrow$), Step 1	`IsDensityOperator.vonNeumannEntropy_eq_zero_iff`
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves_vonNeumannEntropy`, `exists_isWignerMap_of_preserves`
Wigner's theorem, any dimension	`Wigner.IsWignerMap.exists_linearIsometry_or_conjLinearIsometry`, `Wigner.wigner_isometry`
Wigner's theorem, onto the rays	`Wigner.IsWignerMap.exists_unitary_or_antiunitary_of_surjective`
Theorem 2 ($\Leftarrow$), Steps 1–2 with Wigner	`exists_isometry_of_preserves`; `exists_unitary_of_preserves` if every pure state is an image
Theorem 2 ($\Leftarrow$), surjectivity from the reconstructed past	`exists_proj_of_past`, `exists_unitary_of_preserves_of_past`
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`eventually_exists_unitary_of_preserves` (from `Wigner.eventually_exists_unitary_of_unitary_or_antiunitary`)
Theorem 2 ($\Leftarrow$), surjectivity is needed	`shift_preserves_not_unitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)\psi$	`Wigner.exists_unitary_differentiableAt_apply`
Theorem 2 ($\Leftarrow$), Step 3	`ContinuousLinearMap.exists_hasDerivAt_apply`, `star_eq_neg_of_hasDerivAt_apply_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves_vonNeumannEntropy`

Differences from the brief.

- Wigner's theorem in any dimension follows Gehér's proof with a Hilbert basis (b_k). The first normalization changes: the images $f(b_k)$ are orthonormal but need not span, so they define an isometry V with $Vb_k = f(b_k)$, not a unitary. By Parseval, $\|V^\dagger f(v)\|^2 = \sum_k |\langle f(b_k)|f(v)\rangle|^2 = \sum_k |\langle b_k|v\rangle|^2 = 1 = \|f(v)\|^2$, so $f(v)$ lies in the range of V and $V^\dagger \circ f$ is a Wigner map that fixes every b_k. The diagonal unitary and the conjugation are then built from the basis as in finite dimension. The result is a linear or conjugate-linear isometry; it is onto, hence unitary or anti-unitary, when every ray is the image of a ray, and always in finite dimension.

- The anti-unitary branch is excluded by continuity without a trace. For $P = |a\rangle\langle a|$, Q, R rank-one projections onto a, b, c and a fixed vector x, $\langle x|PQRP\,x\rangle = |\langle a|x\rangle|^2\,\Delta(a,b,c)$ (`inner_rankOne_mul_rankOne_mul_rankOne_mul_rankOne_apply`). Take the test triple $b_i, (b_i - b_k)/\sqrt{2}, (b_i - ib_k)/\sqrt{2}$ with $\Delta = (1+i)/4$, $x = b_i$, and the images $\Phi_t(P)$. The imaginary part is $\frac{1}{4}$ at t_0 and continuous at t_0 in operator norm, but ≤ 0 for an anti-unitary, since its triple product is $\overline{\Delta}$ and the factor $|\langle a|x\rangle|^2$ is nonnegative. The finite-dimensional `Wigner.eventually_exists_unitary`, which used $\mathrm{tr}(PQR) = \Delta$, is now a corollary. Continuity is assumed in operator norm; continuity in trace norm implies it.

- EVDIFF is formalized as in finite dimension: $t \mapsto \Phi_t(\rho)$ is differentiable at t_0 in operator norm for every density operator ρ. The conclusion is the Schrödinger equation on vectors, as in DR-SCEQ: $\psi(t) = U(t)\psi$ satisfies $\psi'(t_0) = (i\hbar)^{-1}H\psi(t_0)$ for every ψ, with one bounded self-adjoint H. It is not the equation for $t \mapsto U(t)$ in operator norm, which the finite-dimensional `schrodinger_of_preserves` gives. The basis sum that fixes the phase in finite dimension does not converge in operator norm; instead, for a fixed unit vector e, Φ_t extended by polarization gives $(|Ux\rangle\langle Ue|)\,e = \langle Ue|e\rangle\,Ux$, and dividing by $|\langle Ue|e\rangle|$ fixes the phase one vector at a time. So $t \mapsto U(t)x$ is differentiable at t_0 for each x. By Banach–Steinhaus, applied to the difference quotients, the derivatives $x \mapsto (U(\cdot)x)'(t_0)$ form a bounded operator A. Differentiating $\langle U(t)x|U(t)y\rangle = \langle x|y\rangle$ gives $A^\dagger = -A$, and $H = i\hbar A$.

- No spectral decomposition is used. The entropy is the trace of $\eta(\rho)$, and pure states are found through the spectrum: $S(\rho) = 0$ forces $\eta(\rho) = 0$, so η vanishes on the spectrum, which then lies in $\{0,1\}$; ρ is idempotent, and an idempotent density operator is $|\psi\rangle\langle\psi|$ (`IsDensityOperator.eq_proj_of_isIdempotentElem`).

- Corollary 1 uses the cubic relation as in finite dimension, with the spectral mapping theorem for polynomials in place of eigenvalues. Then $\eta(\rho) = \alpha\rho + \beta\rho^2$, and its trace is computed from $\sum_i \langle e_i|x\rangle\langle y|e_i\rangle = \langle y|x\rangle$ for the rank-one terms.

- The $(\Rightarrow)$ direction uses only $U(t)^\dagger U(t) = 1$, which every propagator satisfies. Conjugation $X \mapsto UXU^\dagger$ by an isometry is a non-unital star algebra homomorphism, so it commutes with the functional calculus of η (since $\eta(0) = 0$), and it preserves `posTrace`. Unitarity is not needed.

- Entropies are compared in $[0, \infty]$; in infinite dimension $S(\rho) = \infty$ occurs.

- DR-INFO is formalized in full: $S(\Phi\rho) = S(\rho)$, and every density operator ρ has a reconstructed past, a density operator σ with $\Phi\sigma = \rho$; then $S(\sigma) = S(\rho)$ by the first condition. The finite-dimensional part does not need the past, and `exists_isometry_of_preserves` shows what remains without it.

Problems found in the brief.

- In infinite dimension the reconstructed past in DR-INFO is essential, and Step 1 uses it when it says that pure states "correspond bijectively". DR-INFO presupposes a past ensemble ρ_P for every initial ensemble, so Φ is onto the density operators, as in the introduction's "deterministic and reversible". Then every pure state is the image of a pure state, since its past has entropy zero (`exists_proj_of_past`), and Wigner's theorem gives a unitary or an anti-unitary (`exists_unitary_of_preserves_of_past`). The brief's proof is therefore correct, but the past cannot be dropped. Conjugation $\rho \mapsto S\rho S^\dagger$ by the shift S on $\ell^2(\mathbb{N})$ maps density operators to density operators and preserves the entropy and equal mixtures, but no pure state goes to $|e_0\rangle\langle e_0|$. So it is induced by no unitary and by no anti-unitary (`shift_preserves_not_unitary`), and Wigner's theorem gives only an isometry. In finite dimension the past is automatic.

- The time structure of [B]EVMIX does not replace the past. On $L^2(0,\infty)$ the right shifts $(V_t f)(x) = f(x-t)$ for $x > t$ and 0 otherwise are isometries forming a strongly continuous semigroup with $V_0 = I$. Then $\Phi_{t_0\to t_1}(\rho) = V_{t_1-t_0}\rho V_{t_1-t_0}^\dagger$ satisfies [B]EVMIX, preserves the entropy, and is induced by no unitary for $t_1 > t_0$; it has no reconstructed past. (By the Wold–Cooper decomposition, every strongly continuous semigroup of isometries is a unitary group plus copies of this one.) Not formalized. A norm-continuous semigroup of isometries is $V_t = e^{tA}$ with bounded A, and differentiating $V_t^\dagger V_t = I$ at 0 gives $A^\dagger = -A$, so it is unitary. The example therefore needs an unbounded generator, which Step 3 does not cover anyway.

- EVDIFF for every density operator forces a bounded Hamiltonian, as the Banach–Steinhaus step shows. Unbounded Hamiltonians, whose evolution is differentiable only on a dense set of states, are outside the theorem as stated; the brief "Multiple equivalent characterizations of the Schrödinger equation" treats them with Stone's theorem. Otherwise no problems beyond those of the finite-dimensional part.